DATA ANALYTICS
PYTHON ASSIGNMENT 3 : While Loop, For loop and Function
SUBMISSION:11.10.2026
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While Loop, For loop and Function

While loop & Control Statements : (else, break, continue, pass)

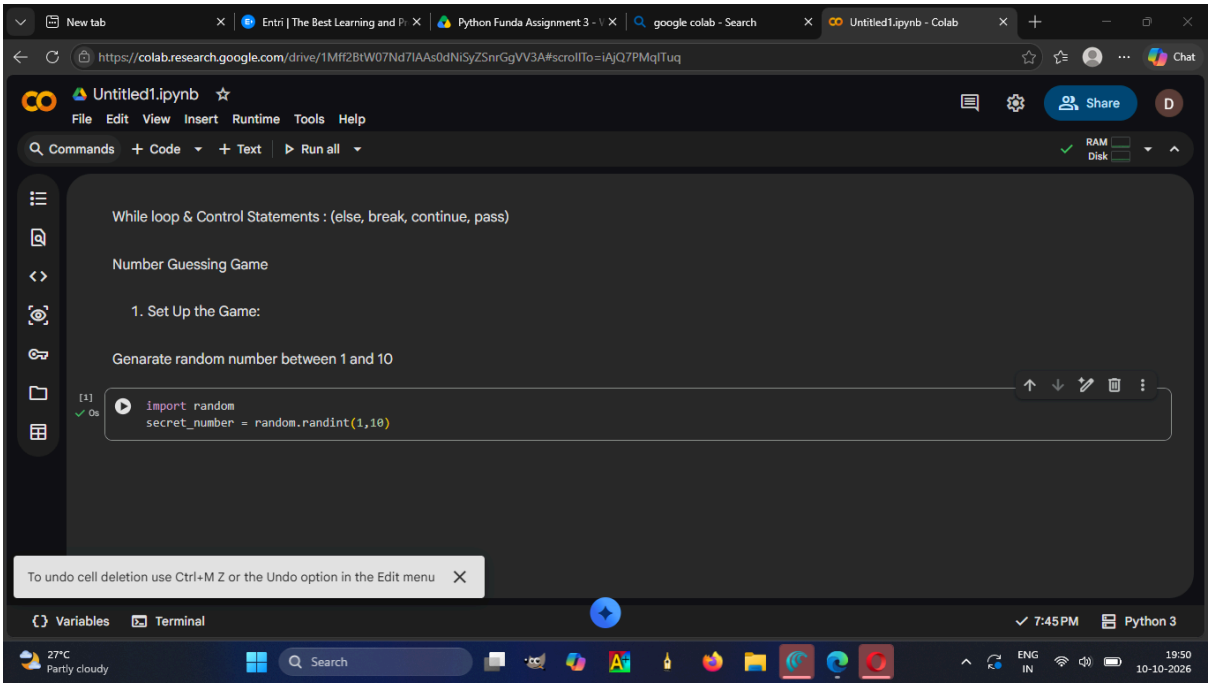
1. Number Guessing Game

Problem Statement: Create a Python program that implements a simple number guessing game using a while loop. The program should make use of control statements such as else, break, and continue.

Task 1 : Set Up the Game: Generate a random number between 1 and 10 that the user has to guess. Import random and use randint function.

Solution :

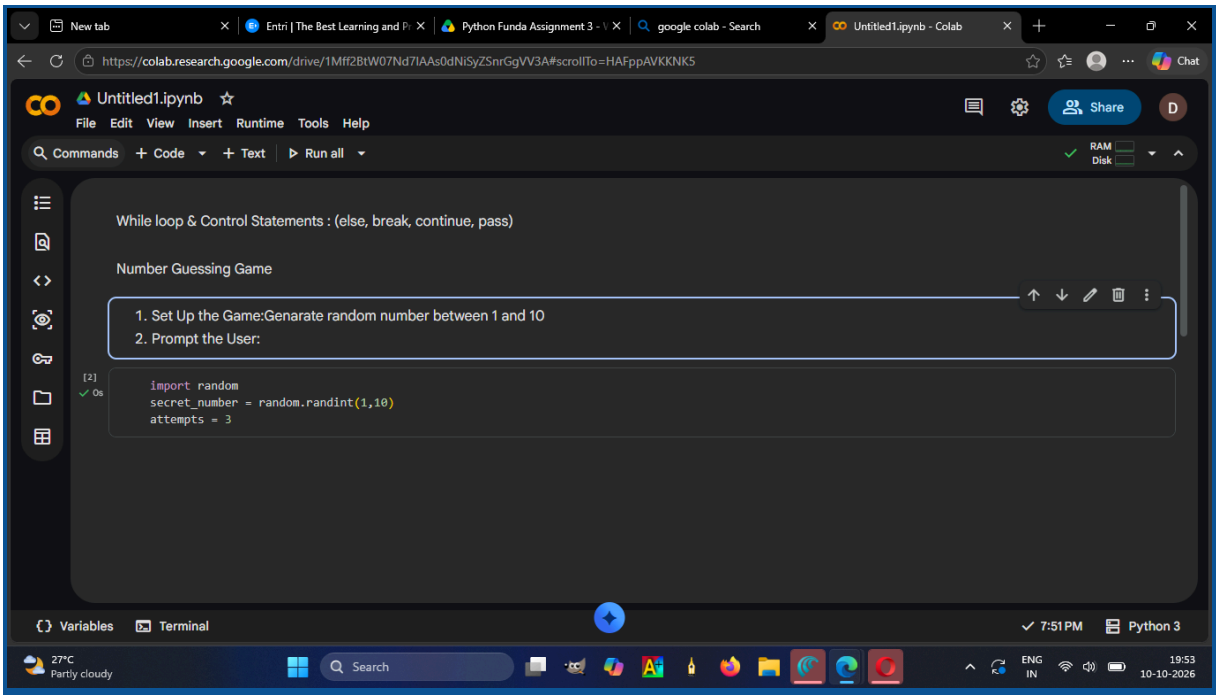
Preview screenshot :



Task 2 : Prompt the User: Ask the user to guess the number and set a variable attempt to 3, which represents the maximum number of guesses allowed.

Solution:

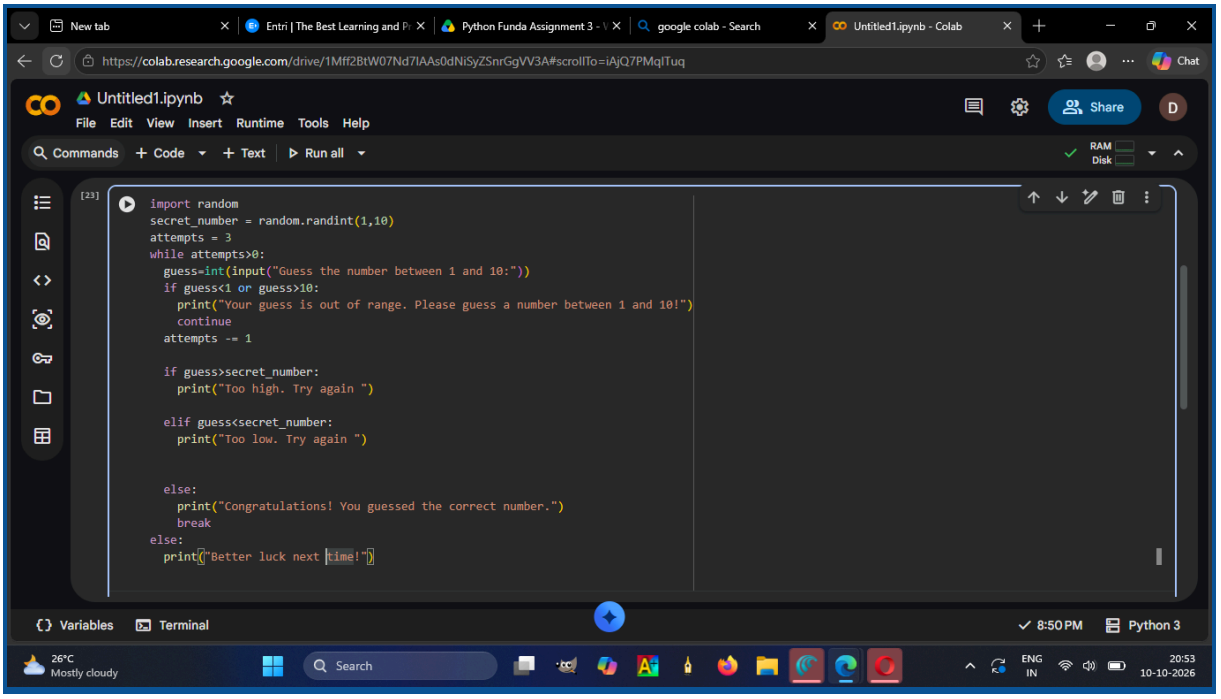
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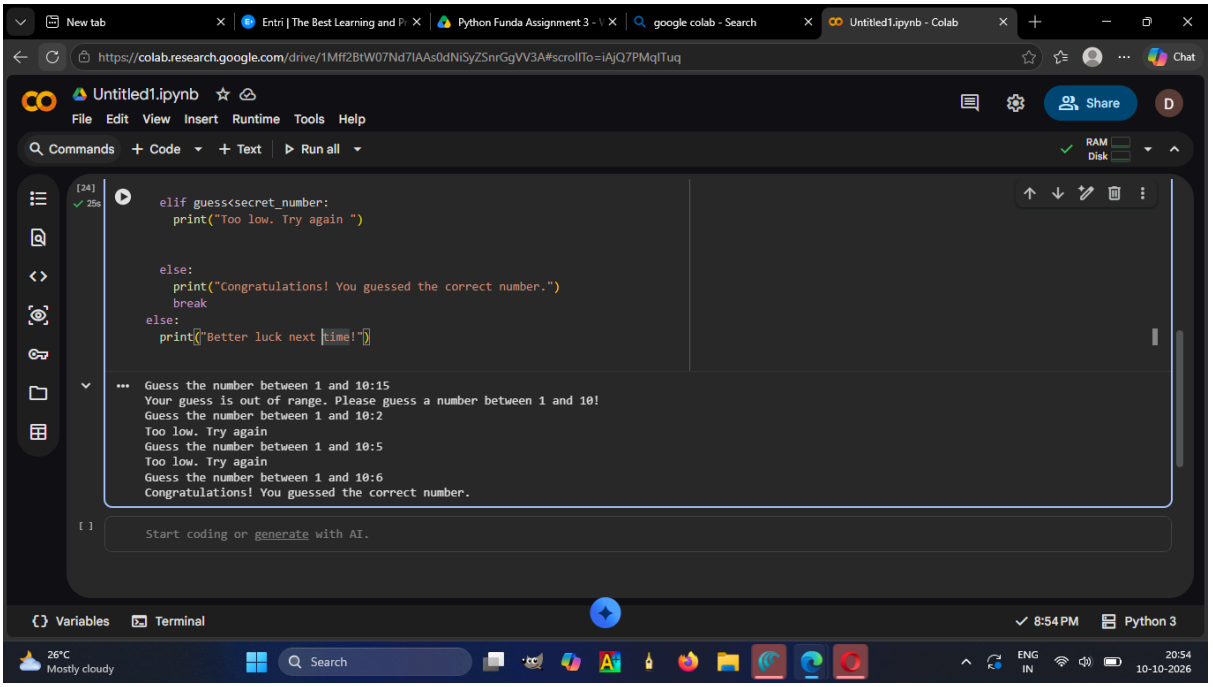


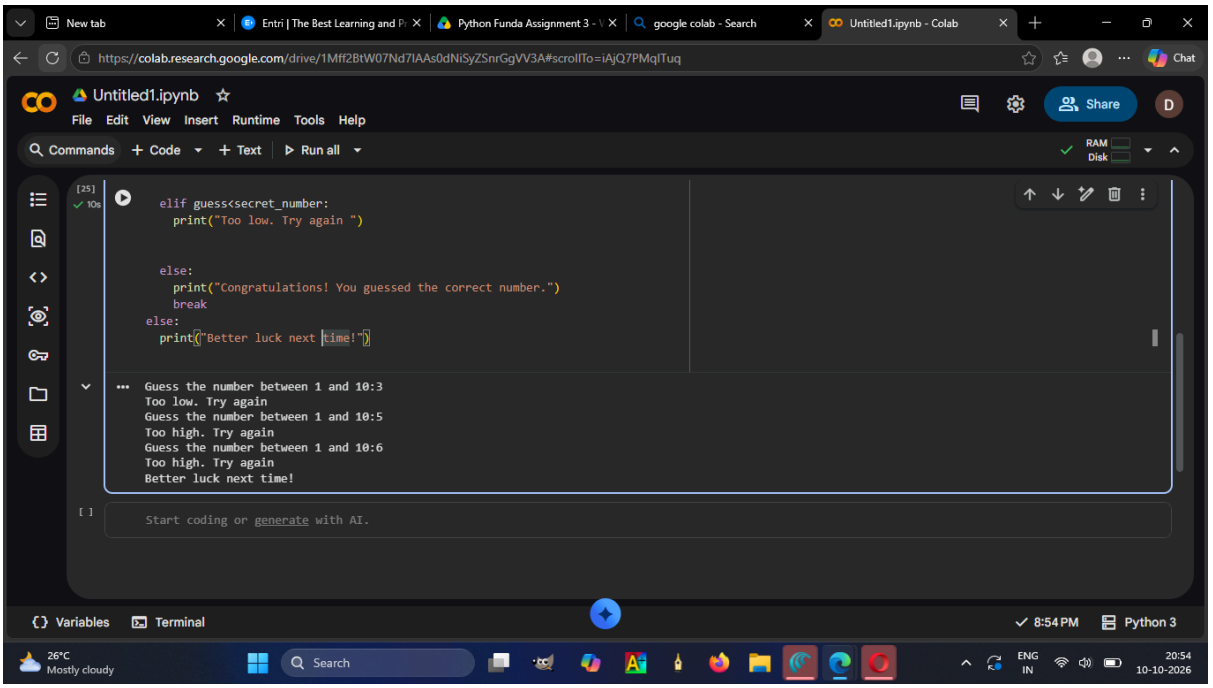
Task 3 : Implement the Guessing Logic by using While Loop

Solution :

Preview screenshot :







Explanation:

- For creating a Python program that implements a simple number guessing game using a while loop , I had used the if , elif and else statement with while loop for implementing the number guessing game.
- Step 1 : For generating a random number between 1 and 10 that the user has to guess. I used the Import random and randint function.

    - Import random - Imports the random module.
    - random.rendit(1,10) - Generate random numbers between 1 and 10.

- Step 2:I had set a variable attempt to 3, which represents the maximum number of guesses allowed. So I had used the attempt -=1 within the while loop because it will reduce the remaining attempts after an incorrect guess.
- Step 3: I had used a while loop to allow the user to keep guessing until they get the correct number or run out of attempts.

    - First while checking the attempt>0 then it will run the loop and will ask the used number between 1 and 10.
    - Then the “IF” statement will check if the user input is less than 1 or greater than 10. If the guess is out of the valid range (1 to 10), displays “Your Guess is out of range” and continues the game with the “Continue” statement.
    - If the “IF” condition fails it will reduce the attempt and run the next “IF” statement and check the input is greater than the secret number ,if it fails it will run the elif condition or else it will display “Too high try again “.
    - If the “IF” condition fails it will run elif and “Elif” will check the input is less then secret number, if it fails it will run the “else” condition or else it will display “Too Low try again “.
    - If the “Elif” condition fails it will run “Else” and congratulate the user and end the game with a ” break” statement.
    - If the user runs out of attempts without guessing the correct number .It will display “"Better luck next time!" .
    - The above mentioned screenshots represent the results.

For Loop:

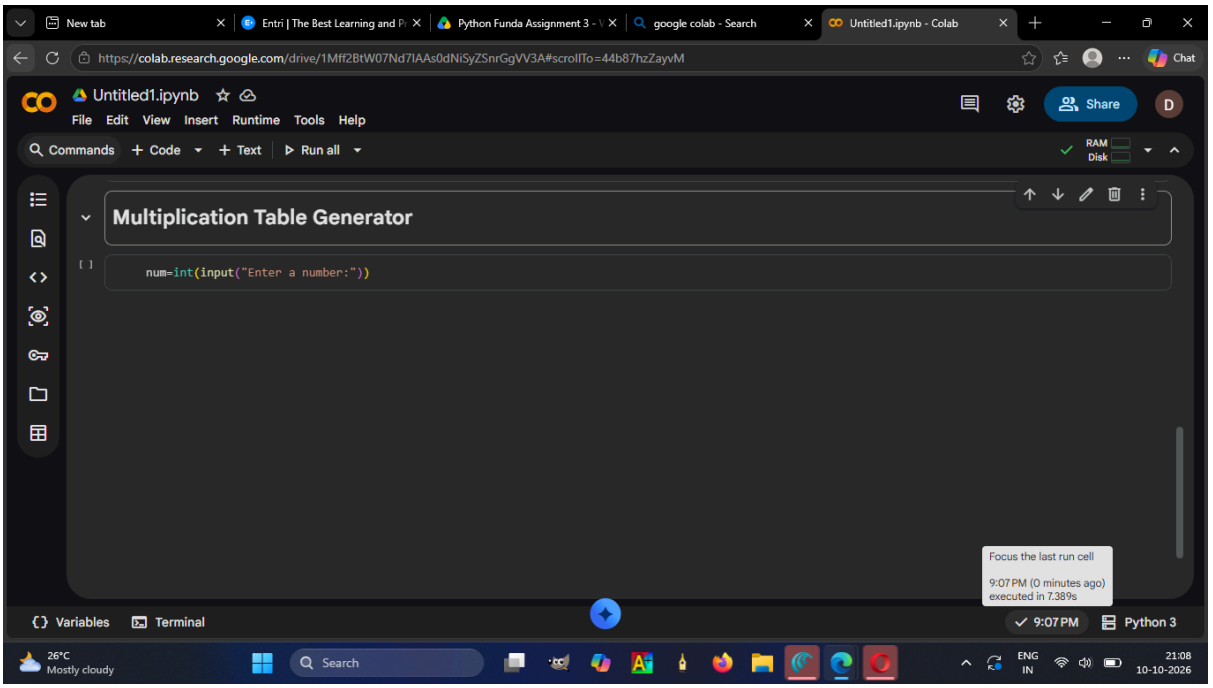
Multiplication Table Generator

Problem Statement: Create a Python program that generates and prints a multiplication table (from 1 to 10) for a given number using a for loop and the range function.

Task 1 : Prompt user for Input. Ask the user to enter a number for which they want to generate a multiplication table.

Solution :

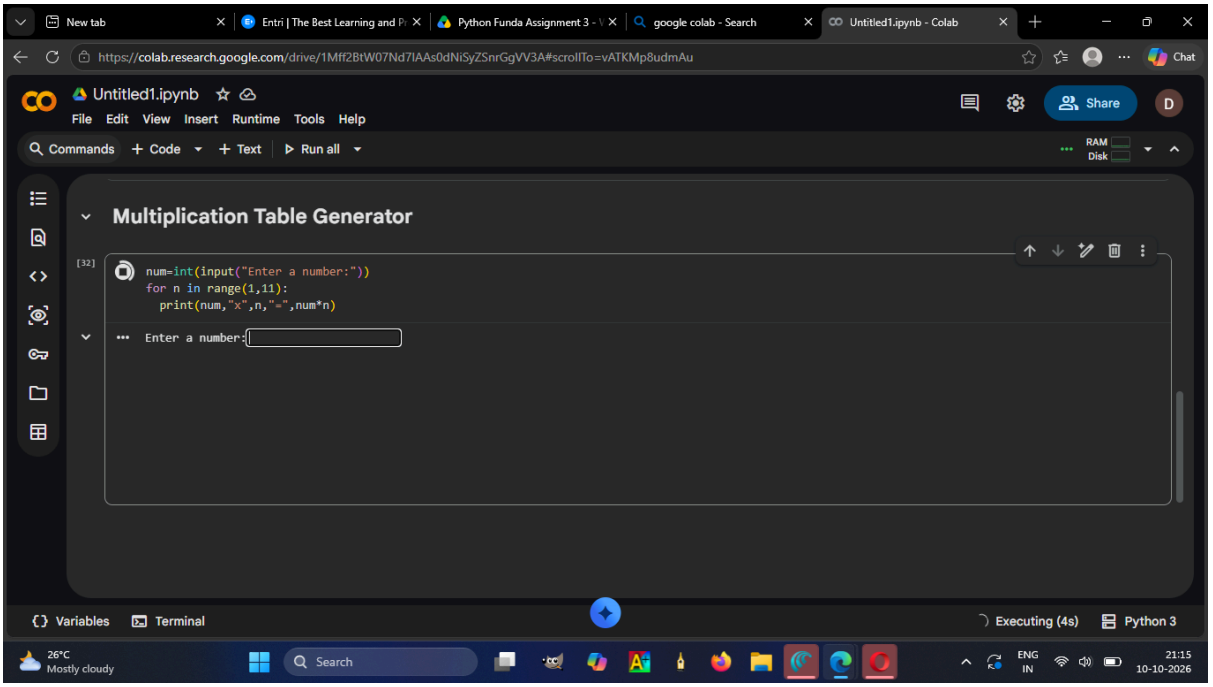
Preview screenshot :



Task 2 : Generate the Multiplication Table: Use a for loop to iterate through the numbers 1 to 10. In each iteration, calculate the product of the user's number and the current number from the loop.

Solution :

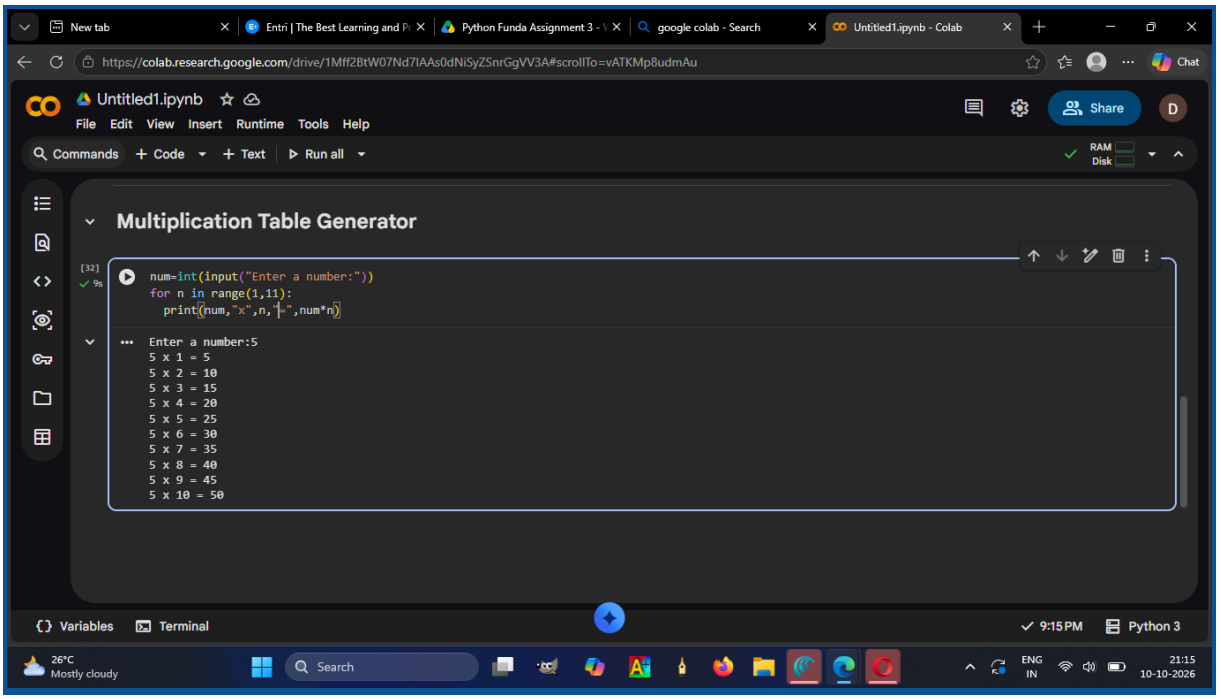
Preview screenshot :



Task 3 : Display the Multiplication Table: Print each line of the multiplication table in the format: "number x i = result".

Solution :

Preview screenshot :



Explanation:

- For creating a multiplication table generator I had used the “FOR LOOP”.
- First I had to create a variable name “num” in integer type with input function. It will ask the user to enter a number for which they want to generate a multiplication table.
- I had to write the code by “FOR LOOP” to generate the Multiplication Table by using a for loop to iterate through the numbers 1 to 10. I had created this code (for n in range(1,11):) to calculate the product of the user's number and the current number from the loop .
- Then I had create the code print(num,"x",n,"=",num*n) for printing the result .
- The above mentioned screenshots represent the result.

Function:

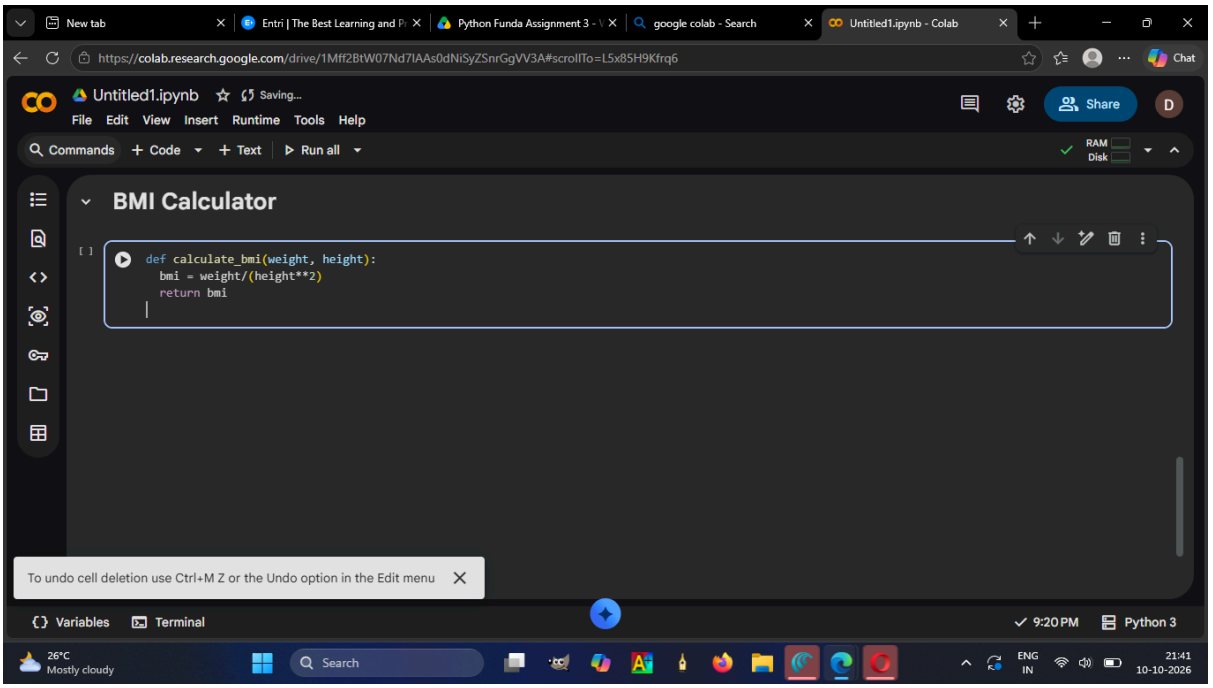
BMI Calculator

Problem Statement: Create a Python program that calculates the Body Mass Index (BMI).

Task 1 : Define a function calculate_bmi(weight, height) that returns the BMI.

Solution :

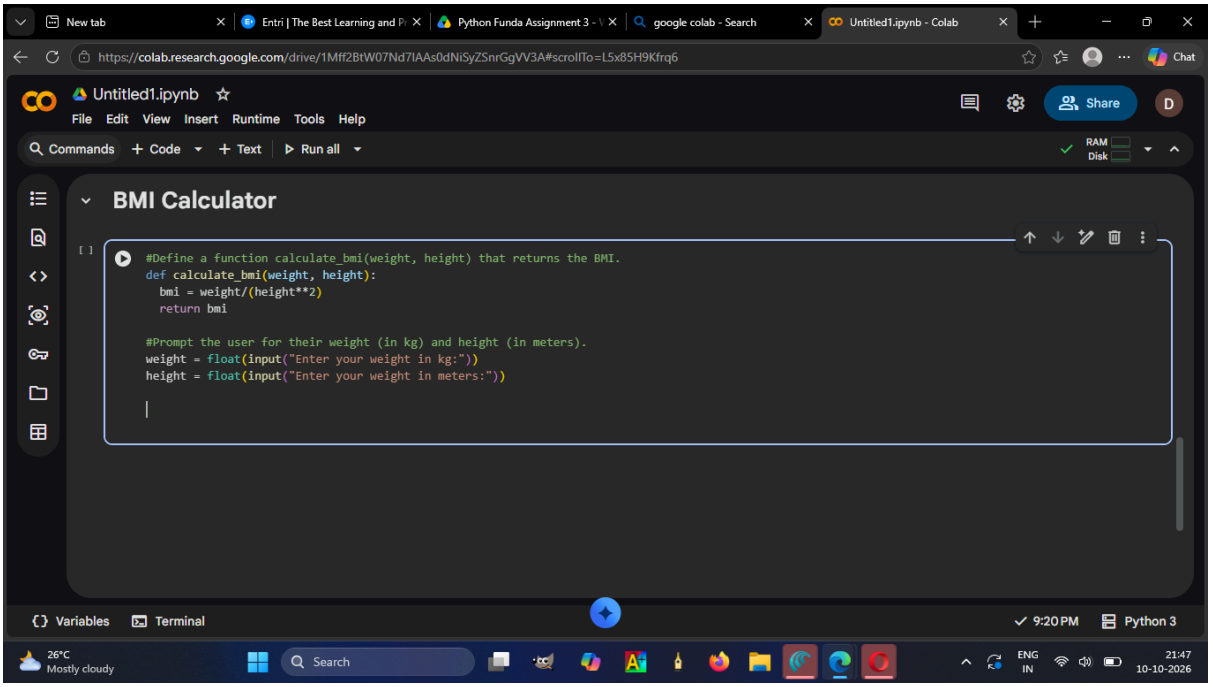
Preview screenshot :



Task 2 : Prompt the user for their weight (in kg) and height (in meters).

Solution :

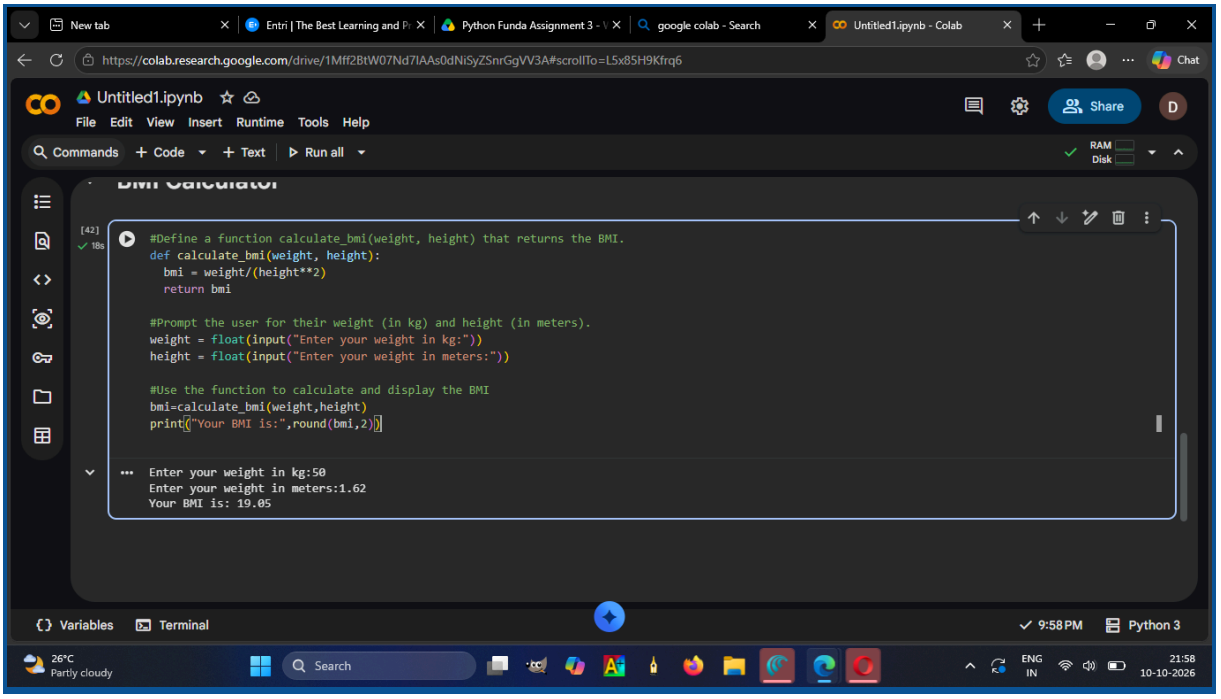
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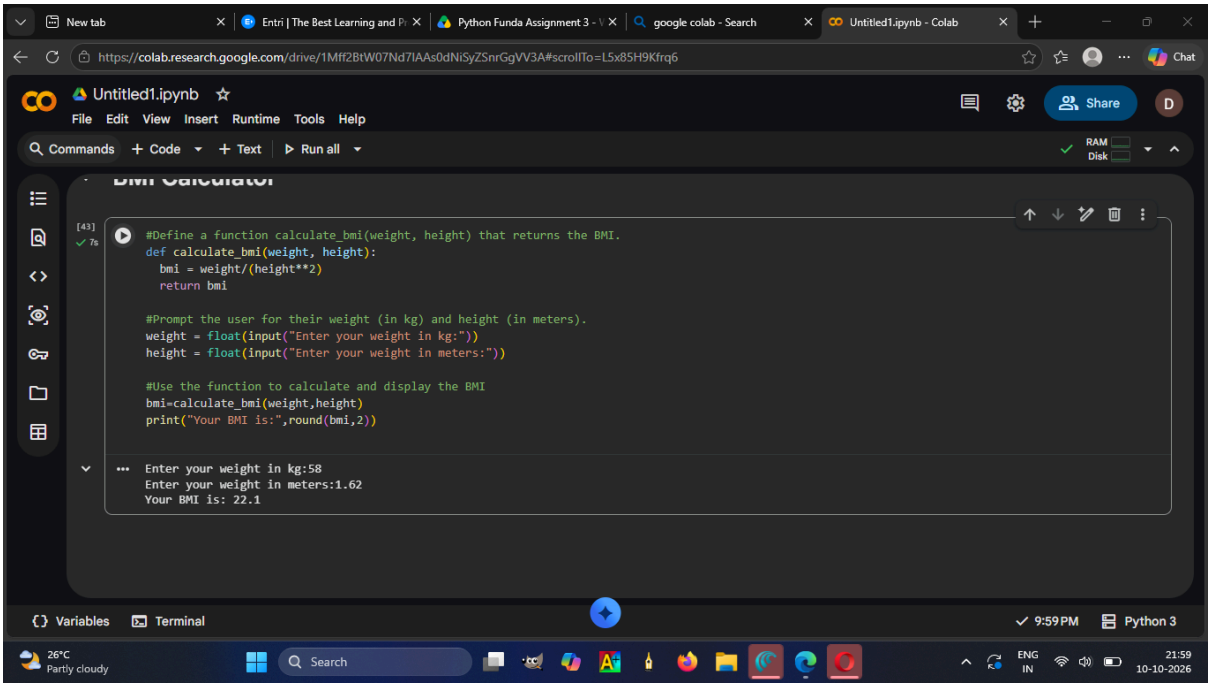


Task 3 : Use the function to calculate and display the BMI

Solution :

Preview screenshot :





Explanation :

- For creating BMI Calculator I had to define a function calculate_bmi(weight, height) , (bmi = weight/(height**2))that returns the BMI by using the “def” keyword .
- Then write code to get input from users as floating type for their weight (in kg) and height (in meters).
- Then use the function to calculate and display the BMI (bmi=calculate_bmi(weight,height)).
- The above mentioned screenshots represent the result.